J-C-115a
September 6, 1962

SUPERSEDING
Int. Fed. Spec. J-C-00115 (GSA-FSS)
September 18, 1961

FEDERAL SPECIFICATION

CABLE AND WIRE, POWER, ELECTRICAL, (SERVICE-ENTRANCE AND SERVICE-DROP)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers service-entrance and service-drop cables and wires employing copper or aluminum, solid or stranded conductors for use on circuits involving potentials of 600 volts or less, and on circuits involving potentials of 300 volts or less to ground in the case of cables having an uninsulated conductor.

1.2 Classification.

1.2.1 *Types, grades, styles, and classes.* Cable and wire furnished under this specification shall be of the following types, grades, styles, and classes, as specified (see 6.1).

Types:

Type SE.—For use in dry or wet locations other than underground.

Type USE.—For direct burial in earth and having no inherent protection against mechanical abuse.

Grades:

Grade I.—With RHW grade rubber insulation.

Grade II.—With TW grade thermoplastic insulation (Type SE only).

Grade III.—With THW grade thermoplastic insulation (Type SE only).

Styles (Type SE only):

Style A.—With flat metal armor for mechanical protection.

Style U.—Without metal armor.

Classes:

Class 1.—With uninsulated ground conductor.

Class 2.—Without grounding conductor.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

J-C-98—Cable and Wire, Insulated; Methods of Sampling and Testing.
UU-P-271—Paper, Wrapping, Waterproofed Kraft.
PPP-B-601—Boxes, Wood, Cleated-Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Box, Fiberboard.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Docu-

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ments, U.S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL-L-10547—Liner, Case, Waterproof.

Military Standards:

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification.

Underwriters' Laboratories, Inc., Publication:

UL 854—Standard For Service Cables.

(Copies may be obtained upon application to their addresses: 161 Sixth Avenue, New York, N. Y.; 207 East Ohio Street, Chicago 11, Ill.; 1655 Scott Blvd., Santa Clara, Cal.).

3. REQUIREMENTS

3.1 Cable and wire covered by this specification shall conform to the requirements of Underwriters' Laboratories, Inc. Standard UL 854 for Service Cables.

3.2 **Fire and casualty hazards.**

3.2.1 Each bidder shall submit to the contracting agency proof that the cable he proposes to supply under this specification conforms to the requirements of UL 854, Standard for Service Cables. The label or listing of the Underwriters' Laboratories, Inc. may be accepted as evidence that the cable conforms to these requirements.

3.2.2 In lieu of the Underwriters' Laboratories, Inc. label or listing, the bidder may submit independent proof satisfactory to the contracting agency that his cable conforms to the published standards as determined by inspection and tests specified in the above referenced UL 854, Standard for Service Cables.

3.2.3 Compliance with the above preliminary requirements in regard to fire and casualty hazards does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his material.

3.3 **Conductor size.** Cable and wire furnished under this specification shall be of the following conductor sizes (see 6.1).

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Supplier responsibilities for inspection.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Number of Conductors	*Size*	
	Copper	*Aluminum*
1	AWG14—1000 MCM	AWG12—1500 MCM
2	AWG12—3/0	AWG10—4/0
3	do	do
4	do	do

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4.1.1 Unless otherwise specified, such inspections by the supplier may be performed on a statistical or production lot basis to assure that the quality of manufacture is equal to the standards set by Underwriters' Laboratories, Inc. (see 6.2).

4.2 Sampling.

4.2.1 *Lot.* Unless otherwise specified, in the invitation for bids, the lot shall be formed as described in paragraph 3 of section 6 of J-C-98.

4.2.2 *Sampling for inspection and tests.* Unless otherwise specified in the invitation for bids, the samples shall be taken as described in paragraph 6 of section 6 of J-C-98.

4.3 Visual inspection.

4.3.1 Each inspection unit in the sample (4.2.2), as well as the package from which the inspection unit is taken, shall be examined visually before any tests are made. Visual inspection shall cover construction, splices, marking, type of conductor, joints, repairs, and overall workmanship and may be limited to that part of the package which is visible without unrolling.

4.4 Tests.

4.4.1 The cable and wire shall be tested in accordance with UL 854, Standard for Service Cables.

5. PREPARATION FOR DELIVERY

For civil agency procurement, the definitions and applications of the various levels of packaging and packing shall be as specified in Fed. Std. No. 102.

5.1 Packaging.

5.1.1 *Levels A and B*

5.1.1.1 *Coils (AWG sizes 14 to AWG 10).* Cable and wire shall be uniformly coiled and packaged 250 feet in a fiberboard box conforming to PPP-B-636, type I or type II, class 2, provided with the special knock-out feature normally used in commercial practice for dispensing purposes.

5.1.1.2 *Coils (AWG sizes 8 to AWG 2).* Cable and wire shall uniformly coiled and packaged 250 feet per coil. The coils shall be spirally wrapped with waterproofed kraft paper conforming to UU-P-271, class E-2, and sealed.

5.1.1.3 *Reels.* Cable and wire shall be packaged on non-returnable wood or metal spools or reels suitably designed with strong flanges of sufficient diameter to protect the cable or wire from damage during shipment and storage. After the cable or wire has been wound on the reel or spool, it shall be covered with a full wrap of waterproof kraft barrier material conforming to UU-P-271, class E-2. The wrap shall extend between the flanges and be secured in place with a minimum of two flat steel straps.

5.1.2 *Level C.* Cable and wire shall be packaged in coils or on reels or spools in accordance with standard commercial practice, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 *Level A.*

5.2.1.1 *Coils, (AWG Sizes 14 to AWG 10).* Cable and wire packaged as specified in 5.1.1.1 shall be packed in nailed wood or wood-cleated plywood boxes conforming to PPP-B-621 and PPP-B-601, respectively. Containers shall be fitted with waterproof caseliners conforming to MIL-L-10547, class 2 and sealed in accordance with the appendix thereto.

5.2.1.2 *Coils (AWG Sizes 8 to AWG 2) and reels.* Cable and wire packaged as specified in 5.1.1.2 and 5.1.1.3 shall require no further overpacking.

5.2.2 *Level B, coils and reels.* Cable and wire packed as specified in 5.1.1.1, 5.1.1.2 or 5.1.1.3 shall require no further overpacking.

5.2.3 *Level C.* Cable and wire shall be packaged in substantial containers of the type, size, and kind commonly used for the

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purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required in the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, symbol, and date of this specification.

(b) Type, grade, style, and class of cable and wire (1.2.1).

(c) Conductor material.

(d) Number of conductors and conductor size (3.3).

(e) Administrative provisions for inspection records (4.1).

(f) Lot formation, if other than specified (4.2.1).

(g) Sampling, if other than specified (4.2.2).

(h) Level of packaging and packing required (5.1 and 5.2).

(i) Special marking, if required (5.3).

6.2 If laboratory tests by the Government are required, purchasing officers should order wire or cable for test purposes in addition to the number of feet required for installation.

6.3 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Copper

Rail:

Cable, Electric, Copper, Not Otherwise Indexed By Name.
Carload minimum weight 30,000 pounds.

Motor:

Cable, Electric, Copper, Not Otherwise Indexed.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Aluminum

Rail:

Cable, Electric, Aluminum.
Carload minimum weight 30,000 pounds.

Motor:

Cable, Electric, Aluminum.
Truckload minimum weight 30,000 pounds, subject to Rule 34, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

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